

GROUP 1

CUSTOMER CHURN PREDICTION USING DECISION TREE-BASED CLASSIFICATION MODELS

ISP565: Data Mining

Presented by Group 1

OUR TEAM MEMBERS



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TASK A1

Introduction

Customer Churn Prediction using Decision Tree-Based Classification Models focuses on classifying customers as churn or non-churn to support data-driven retention strategies.

Problem Statement

- Many organizations experience high customer churn that affects revenue and profitability.
- Customer data is available but not effectively analyzed to predict churn early.
- Manual churn analysis is inefficient and time-consuming.
- Some predictive models lack interpretability for business decision-making.
- There is a need for an accurate and understandable decision tree-based churn prediction model.

TASK A1

Data Description

Dataset: Customer Churn Dataset

- Source: Telco Customer Churn Dataset
- Instances: ~7,000 customer records
- Attributes: 21 (Categorical + Numerical)
- Data Type: Structured tabular data

Key Features

- Demographics: Gender, Senior Citizen, Partner, Dependents
- Service Information: Phone Service, Internet Service, Streaming Services, Tech Support
- Contract & Billing: Contract Type, Paperless Billing, Payment Method
- Usage & Charges: Tenure, Monthly Charges, Total Charges

Target Variable

- Y – Churn: Customer leaves the service
- N – No Churn: Customer remains with the service

Collected Via

- Customer subscription and billing records
- Historical customer usage and service information

DATA

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	No	No	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No
5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No	No	No	One year	No	Mailed check	56.95	1889.5	No
3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes
7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	No	Yes	Yes	No	No	One year	No	Bank transfer (automatic)	42.3	1840.75	No
9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes	Electronic check	70.7	151.65	Yes
9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to-month	Yes	Electronic check	99.65	820.5	Yes
1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No	No	Yes	No	Month-to-month	Yes	Credit card (automatic)	89.1	1949.4	No
6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	Yes	No	No	No	No	No	Month-to-month	No	Mailed check	29.75	301.9	No
7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes	Month-to-month	Yes	Electronic check	104.8	3046.05	Yes
6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No	No	No	No	One year	No	Bank transfer (automatic)	56.15	3487.95	No
9763-GRSKD	Male	0	Yes	Yes	13	Yes	No	DSL	Yes	No	No	No	No	No	Month-to-month	Yes	Mailed check	49.95	587.45	No
7469-LKBCI	Male	0	No	No	16	Yes	No	No	No internet service	No internet service	No internet service	No internet service	No internet service	No internet service	Two year	No	Credit card (automatic)	18.95	326.8	No
8091-TTVAX	Male	0	Yes	No	58	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	One year	No	Credit card (automatic)	100.35	5681.1	No
0280-XJGEX	Male	0	No	No	49	Yes	Yes	Fiber optic	No	Yes	Yes	No	Yes	Yes	Month-to-month	Yes	Bank transfer (automatic)	103.7	5036.3	Yes
5129-JLPIS	Male	0	No	No	25	Yes	No	Fiber optic	Yes	No	Yes	Yes	Yes	Yes	Month-to-month	Yes	Electronic check	105.5	2686.05	No
3655-SNQYZ	Female	0	Yes	Yes	69	Yes	Yes	Fiber optic	Yes	Yes	Yes	Yes	Yes	Yes	Two year	No	Credit card (automatic)	113.25	7895.15	No
8191-XWSZG	Female	0	No	No	52	Yes	No	No	No internet service	No internet service	No internet service	No internet service	No internet service	No internet service	One year	No	Mailed check	20.65	1022.95	No
9959-WOFKT	Male	0	No	Yes	71	Yes	Yes	Fiber optic	Yes	No	Yes	No	Yes	Yes	Two year	No	Bank transfer (automatic)	106.7	7382.25	No
4190-MFLUW	Female	0	Yes	Yes	10	Yes	No	DSL	No	No	Yes	Yes	No	No	Month-to-month	No	Credit card (automatic)	55.2	528.35	Yes
4183-MYFRB	Female	0	No	No	21	Yes	No	Fiber optic	No	Yes	Yes	No	No	Yes	Month-to-month	Yes	Electronic check	90.05	1862.9	No
8779-QRDMV	Male	1	No	No	1	No	No phone service	DSL	No	No	Yes	No	No	Yes	Month-to-month	Yes	Electronic check	39.65	39.65	Yes
1680-VDCWW	Male	0	Yes	No	12	Yes	No	No	No internet service	No internet service	No internet service	No internet service	No internet service	No internet service	One year	No	Bank transfer (automatic)	19.8	202.25	No
1066-JKSGK	Male	0	No	No	1	Yes	No	No	No internet service	No internet service	No internet service	No internet service	No internet service	No internet service	Month-to-month	No	Mailed check	20.15	20.15	Yes
3638-WEABW	Female	0	Yes	No	58	Yes	Yes	DSL	No	Yes	No	Yes	No	No	Two year	Yes	Credit card (automatic)	59.9	3505.1	No
6322-HRPFA	Male	0	Yes	Yes	49	Yes	No	DSL	Yes	Yes	No	Yes	No	No	Month-to-month	No	Credit card (automatic)	59.6	2970.3	No
6865-JZNKO	Female	0	No	No	30	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to-month	Yes	Bank transfer (automatic)	55.3	1530.6	No
6467-CHFZW	Male	0	Yes	Yes	47	Yes	Yes	Fiber optic	No	Yes	No	No	Yes	Yes	Month-to-month	Yes	Electronic check	99.35	4749.15	Yes
8665-UTDHz	Male	0	Yes	Yes	1	No	No phone service	DSL	No	Yes	No	No	No	No	Month-to-month	No	Electronic check	30.2	30.2	Yes
5248-YGIJN	Male	0	Yes	No	72	Yes	Yes	DSL	Yes	Yes	Yes	Yes	Yes	Yes	Two year	Yes	Credit card (automatic)	90.25	6369.45	No
8773-HHUOZ	Female	0	No	Yes	17	Yes	No	DSL	No	No	No	No	Yes	Yes	Month-to-month	Yes	Mailed check	64.7	1093.1	Yes
3841-NFECX	Female	1	Yes	No	71	Yes	Yes	Fiber optic	Yes	Yes	Yes	Yes	No	No	Two year	Yes	Credit card (automatic)	96.35	6766.95	No

TASK A2

Data Preparation And Cleaning: Numeric Attribute Analysis

No.	Name
1	☐ customerID
2	☐ gender
3	☐ SeniorCitizen
4	☐ Partner
5	☐ Dependents
6	☒ tenure
7	☐ PhoneService
8	☐ MultipleLines
9	☐ InternetService
10	☐ OnlineSecurity
11	☐ OnlineBackup
12	☐ DeviceProtection
13	☐ TechSupport
14	☐ StreamingTV
15	☐ StreamingMovies
16	☐ Contract
17	☐ PaperlessBilling
18	☐ PaymentMethod
19	☒ MonthlyCharges
20	☒ TotalCharges
21	☐ Churn

The numeric attributes selected in this dataset represent customer subscription duration and billing behavior.

- These variables are important for identifying usage patterns and understanding their influence on customer churn.
- Customers with shorter tenure or higher charges may show a higher likelihood of churning, making these attributes highly relevant for churn prediction.

TASK A2

Data Preparation And Cleaning: Numeric Attribute Analysis

Selected attribute	
Name: tenure	
Missing: 0 (0%)	
Distinct: 73	
Type: Numeric	
Unique: 0 (0%)	
Statistic	Value
Minimum	0
Maximum	72
Mean	32.371
StdDev	24.559

tenure

Selected attribute	
Name: MonthlyCharges	
Missing: 0 (0%)	
Distinct: 1585	
Type: Numeric	
Unique: 381 (5%)	
Statistic	Value
Minimum	18.25
Maximum	118.75
Mean	64.762
StdDev	30.09

MonthlyCharges

Selected attribute	
Name: TotalCharges	
Missing: 11 (0%)	
Distinct: 6530	
Type: Numeric	
Unique: 6172 (88%)	
Statistic	Value
Minimum	18.8
Maximum	8684.8
Mean	2283.3
StdDev	2266.771

TotalCharges

TASK A2

Data Preparation And Cleaning: Nominal Attribute Analysis

No.	Name
1	☐ customerID
2	☒ gender
3	☒ SeniorCitizen
4	☒ Partner
5	☒ Dependents
6	☐ tenure
7	☒ PhoneService
8	☒ MultipleLines
9	☒ InternetService
10	☒ OnlineSecurity
11	☒ OnlineBackup
12	☒ DeviceProtection
13	☒ TechSupport
14	☒ StreamingTV
15	☒ StreamingMovies
16	☒ Contract
17	☒ PaperlessBilling
18	☒ PaymentMethod
19	☐ MonthlyCharges
20	☐ TotalCharges
21	☐ Churn

The nominal attributes in this dataset represent categorical information related to customer demographics, service subscriptions, contract details, and payment preferences.

- These attributes do not have numerical magnitude but play an important role in describing customer behavior and service usage patterns.
- Nominal data helps the classification model identify patterns and relationships between customer characteristics and churn outcomes.

TASK A2

Data Preparation And Cleaning: Nominal Attribute Analysis

SeniorCitizen

Selected attribute			
Name: SeniorCitizen		Type: Nominal	
Missing: 0 (0%)		Distinct: 2	Unique: 0 (0%)
No.	Label	Count	Weight
1	0	5901	5901
2	1	1142	1142

Partner

Selected attribute			
Name: Partner		Type: Nominal	
Missing: 0 (0%)		Distinct: 2	Unique: 0 (0%)
No.	Label	Count	Weight
1	Yes	3402	3402
2	No	3641	3641

gender

Selected attribute			
Name: gender		Type: Nominal	
Missing: 0 (0%)		Distinct: 2	Unique: 0 (0%)
No.	Label	Count	Weight
1	Female	3488	3488
2	Male	3555	3555

TASK A2

Data Preparation And Cleaning: Nominal Attribute Analysis

MultipleLines

Selected attribute			
Name: MultipleLines		Type: Nominal	
Missing: 0 (0%)		Unique: 0 (0%)	
Distinct: 3			
No.	Label	Count	Weight
1	No phone service	682	682
2	No	3390	3390
3	Yes	2971	2971

OnlineSecurity

Selected attribute			
Name: OnlineSecurity		Type: Nominal	
Missing: 0 (0%)		Unique: 0 (0%)	
Distinct: 3			
No.	Label	Count	Weight
1	No	3498	3498
2	Yes	2019	2019
3	No internet service	1526	1526

DeviceProtection

Selected attribute			
Name: DeviceProtection		Type: Nominal	
Missing: 0 (0%)		Unique: 0 (0%)	
Distinct: 3			
No.	Label	Count	Weight
1	No	3095	3095
2	Yes	2422	2422
3	No internet service	1526	1526

TASK A2

Data Preparation And Cleaning: Nominal Attribute Analysis

StreamingTV

Selected attribute			
Name: StreamingTV		Type: Nominal	
Missing: 0 (0%)		Distinct: 3	
		Unique: 0 (0%)	
No.	Label	Count	Weight
1	No	2810	2810
2	Yes	2707	2707
3	No internet service	1526	1526

Contract

Selected attribute			
Name: Contract		Type: Nominal	
Missing: 0 (0%)		Distinct: 3	
		Unique: 0 (0%)	
No.	Label	Count	Weight
1	Month-to-month	3875	3875
2	One year	1473	1473
3	Two year	1695	1695

PaperlessBilling

Selected attribute			
Name: PaperlessBilling		Type: Nominal	
Missing: 0 (0%)		Distinct: 2	
		Unique: 0 (0%)	
No.	Label	Count	Weight
1	Yes	4171	4171
2	No	2872	2872

TASK A2

Data Preparation And Cleaning: Nominal Attribute Analysis

PaymentMethod

Selected attribute			
Name: PaymentMethod		Type: Nominal	
Missing: 0 (0%)		Distinct: 4	
		Unique: 0 (0%)	
No.	Label	Count	Weight
1	Electronic check	2365	2365
2	Mailed check	1612	1612
3	Bank transfer (automa...	1544	1544
4	Credit card (automati...	1522	1522

TASK A2

Data Preparation And Cleaning: Missing Value Handling

Replacing missing values is a data preprocessing step used to handle incomplete data. Missing values are filled using suitable methods to improve data quality and ensure accurate and reliable model performance.

Selected attribute	
Name: TotalCharges	Type: Numeric
Missing: 11 (0%)	Distinct: 6530
Unique: 6172 (88%)	
Statistic	Value
Minimum	18.8
Maximum	8684.8
Mean	2283.3
StdDev	2266.771

In the dataset, missing values are identified in the TotalCharges attribute, mainly due to customers with very short tenure or newly registered accounts. Based on the dataset, it contains 6,530 distinct values, with 11 missing values (0%).

Selected attribute	
Name: TotalCharges	Type: Numeric
Missing: 0 (0%)	Distinct: 6531
Unique: 6172 (88%)	
Statistic	Value
Minimum	18.8
Maximum	8684.8
Mean	2283.3
StdDev	2265

After replacing missing values, the TotalCharges attribute contains no missing data (0%), ensuring complete numerical records for analysis. This data is now suitable for accurate churn prediction modeling.

TASK A2

Data Preparation And Cleaning: Outlier Detection

Outlier detection identifies extreme values that differ from most data points, helping improve data quality and prevent unusual values from affecting model performance. **In this case, only numeric attributes has outliers.**

Selected attribute	
Name: tenure	Type: Numeric
Missing: 0 (0%)	Distinct: 73
	Unique: 0 (0%)
Statistic	Value
Minimum	0
Maximum	72
Mean	32.371
StdDev	24.559

tenure

Selected attribute	
Name: MonthlyCharges	Type: Numeric
Missing: 0 (0%)	Distinct: 1585
	Unique: 381 (5%)
Statistic	Value
Minimum	18.25
Maximum	118.75
Mean	64.762
StdDev	30.09

MonthlyCharges

Selected attribute	
Name: TotalCharges	Type: Numeric
Missing: 11 (0%)	Distinct: 6530
	Unique: 6172 (88%)
Statistic	Value
Minimum	18.8
Maximum	8684.8
Mean	2283.3
StdDev	2266.771

TotalCharges

- The analysis of minimum, maximum, mean, and standard deviation values revealed extreme values that fall far from the central distribution.
- Attributes with large value ranges and high standard deviations indicate the presence of potential outliers, which may influence model performance and were therefore identified for further preprocessing.

TASK A2

Data Preparation And Cleaning: Data Selection

Data selection involves choosing relevant attributes for analysis while removing unnecessary ones. In this dataset, the **customerID** attribute was removed because it serves only as a unique identifier and contains no analytical value for predicting customer churn.

No.		Name
1	☒	customerID
2	☐	gender
3	☐	SeniorCitizen
4	☐	Partner
5	☐	Dependents
6	☐	tenure
7	☐	PhoneService
8	☐	MultipleLines
9	☐	InternetService
10	☐	OnlineSecurity
11	☐	OnlineBackup
12	☐	DeviceProtection
13	☐	TechSupport
14	☐	StreamingTV
15	☐	StreamingMovies
16	☐	Contract
17	☐	PaperlessBilling
18	☐	PaymentMethod
19	☐	MonthlyCharges
20	☐	TotalCharges
21	☐	Churn

No.		Name
1	☐	gender
2	☐	SeniorCitizen
3	☐	Partner
4	☐	Dependents
5	☐	tenure
6	☐	PhoneService
7	☐	MultipleLines
8	☐	InternetService
9	☐	OnlineSecurity
10	☐	OnlineBackup
11	☐	DeviceProtection
12	☐	TechSupport
13	☐	StreamingTV
14	☐	StreamingMovies
15	☐	Contract
16	☐	PaperlessBilling
17	☐	PaymentMethod
18	☐	MonthlyCharges
19	☐	TotalCharges
20	☒	Churn

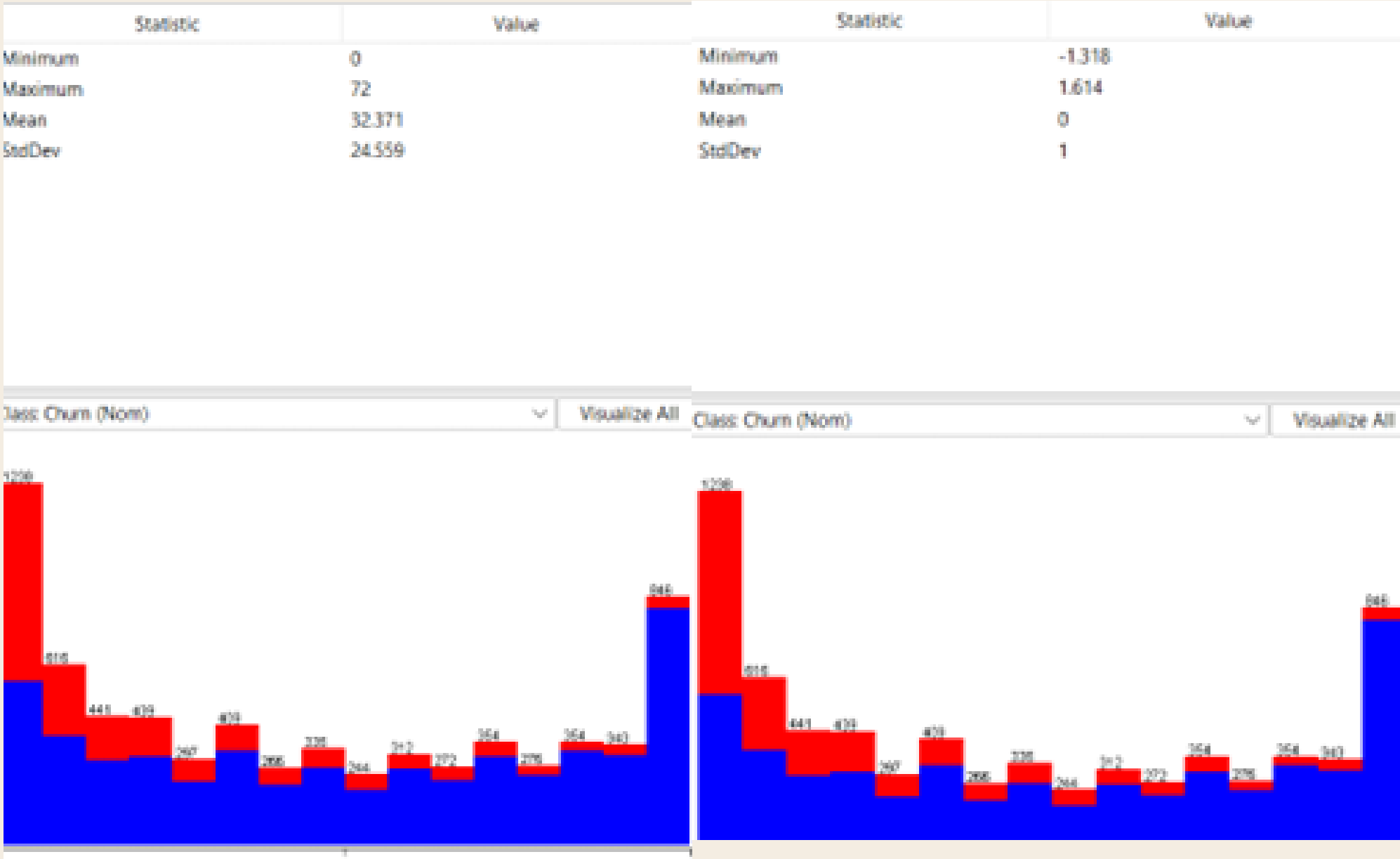
TASK A3 DATA PREPARATION

Filter chosed: Z-score Normalization

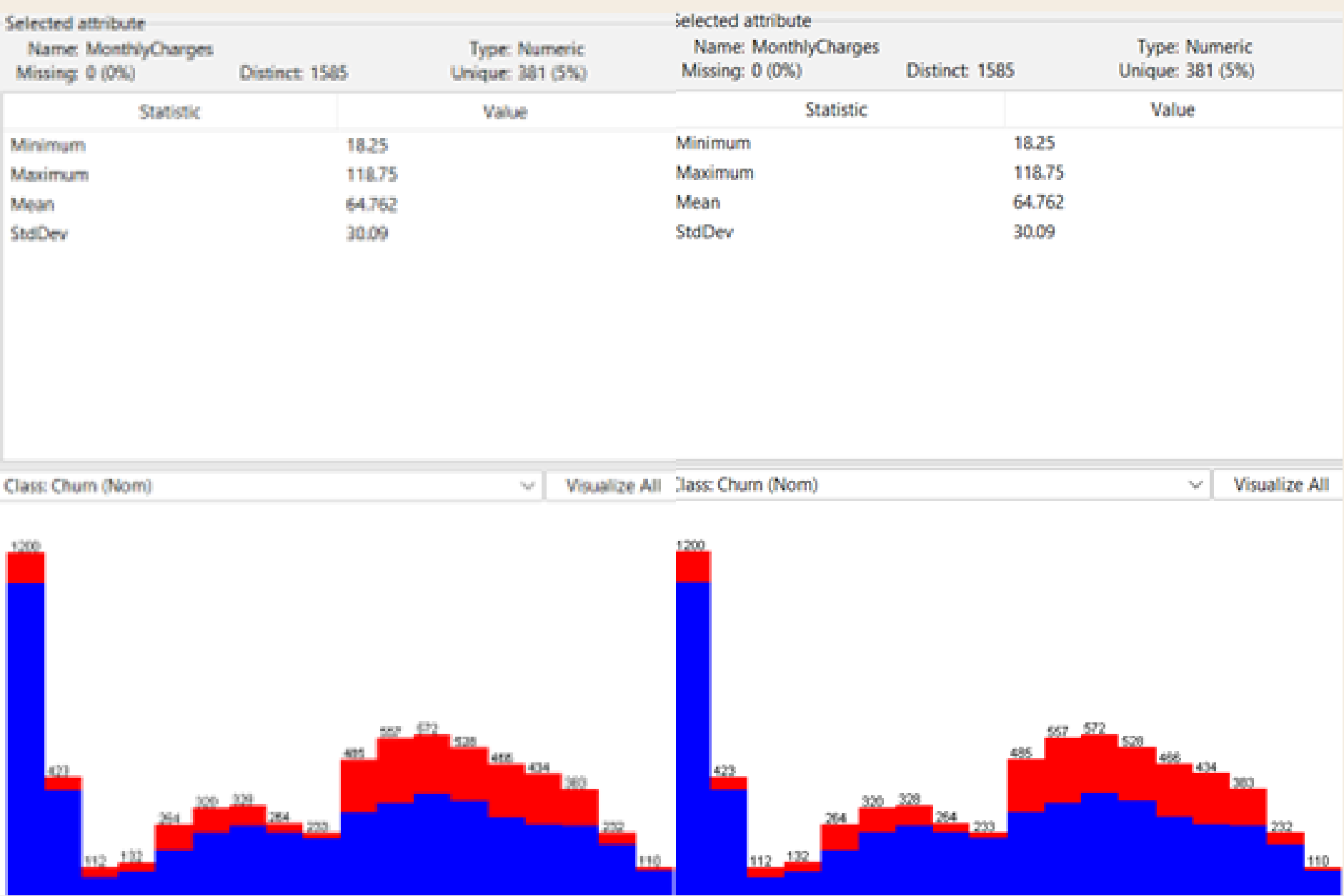
Z-score normalization was applied to numeric attributes using the Standardize filter in WEKA. This method transforms data to have a mean of zero and a standard deviation of one. It was selected because the numeric attributes have different scales and contain extreme values, and Z-score normalization is less sensitive to outliers compared to min-max normalization.

Before

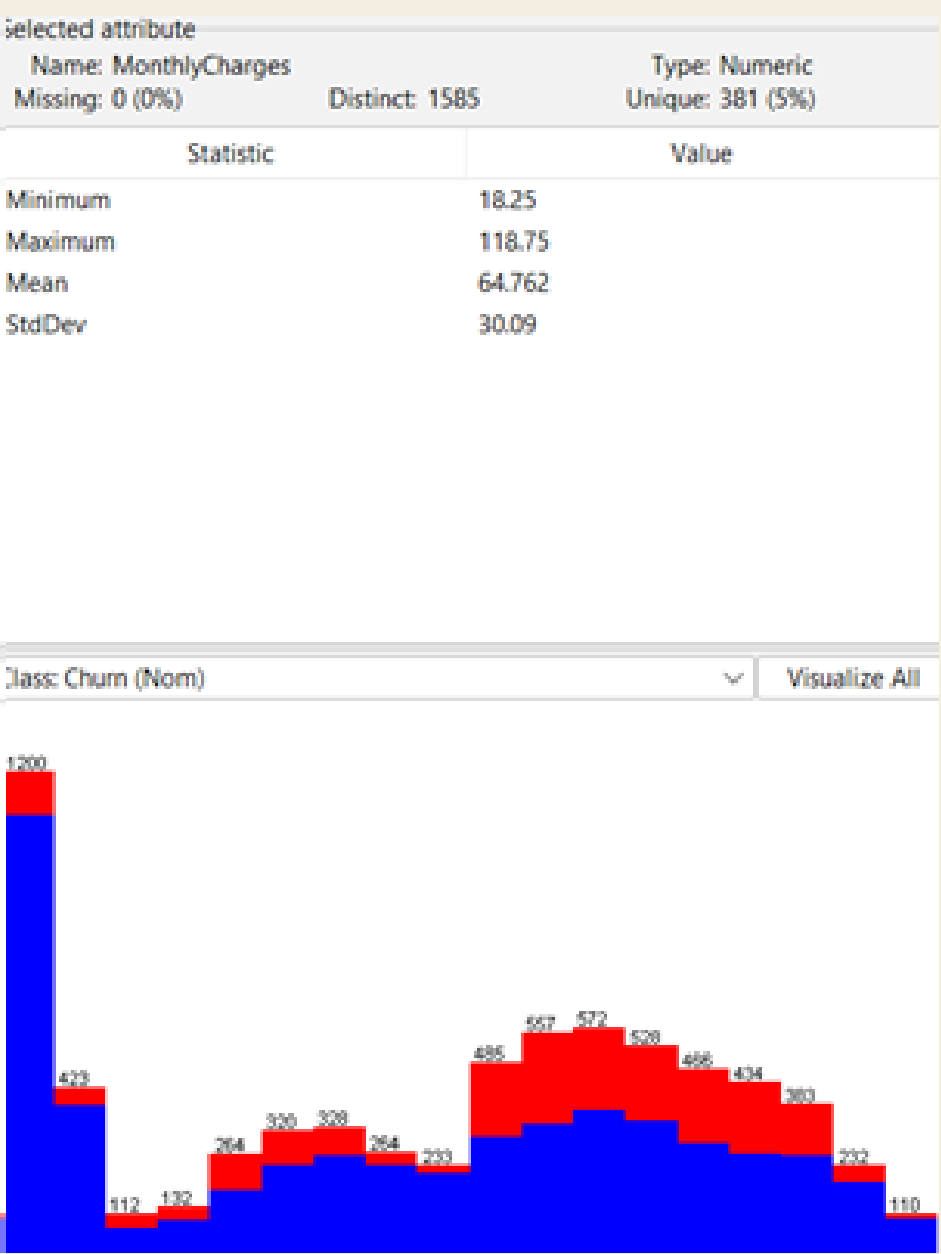
After



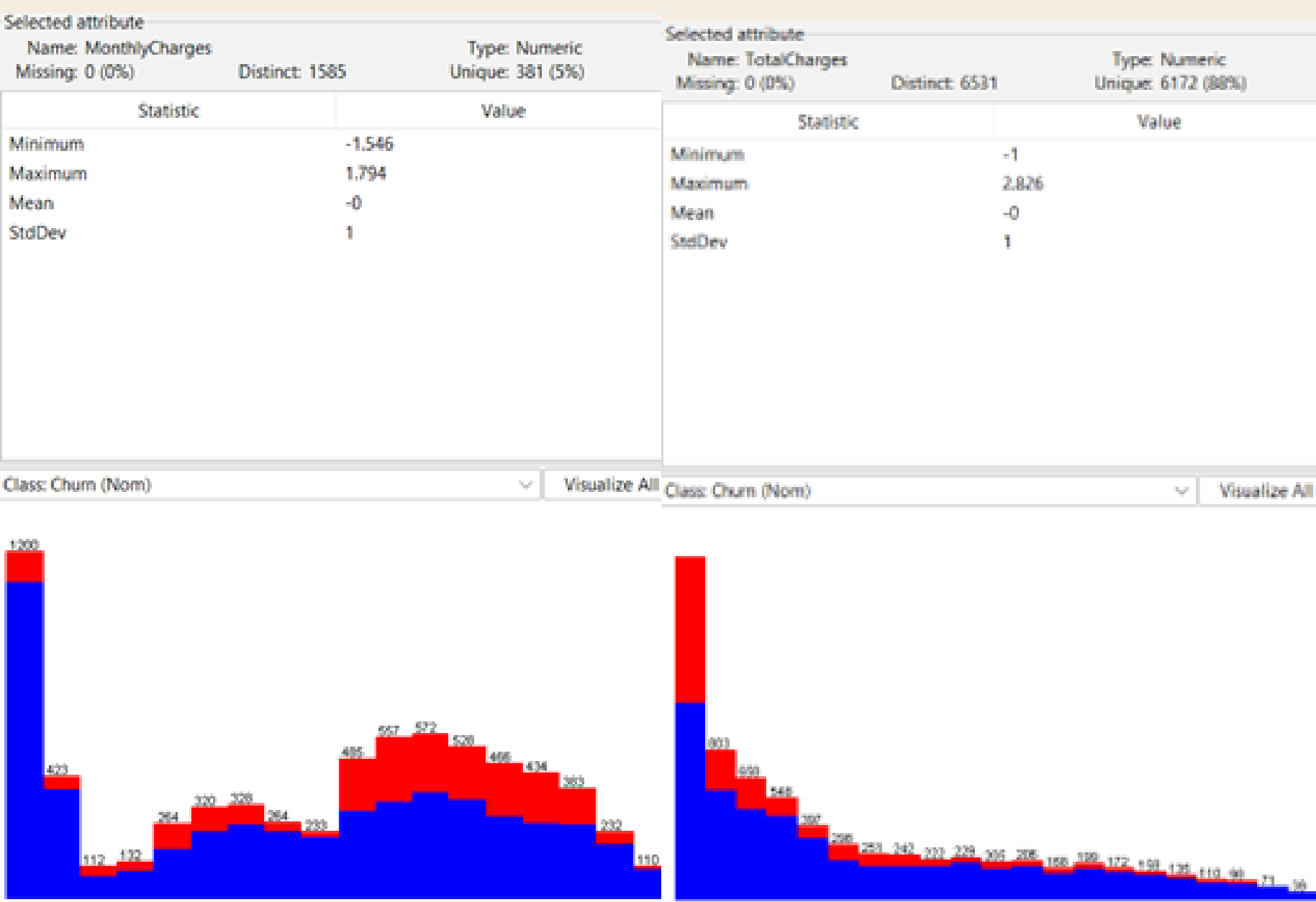
Before



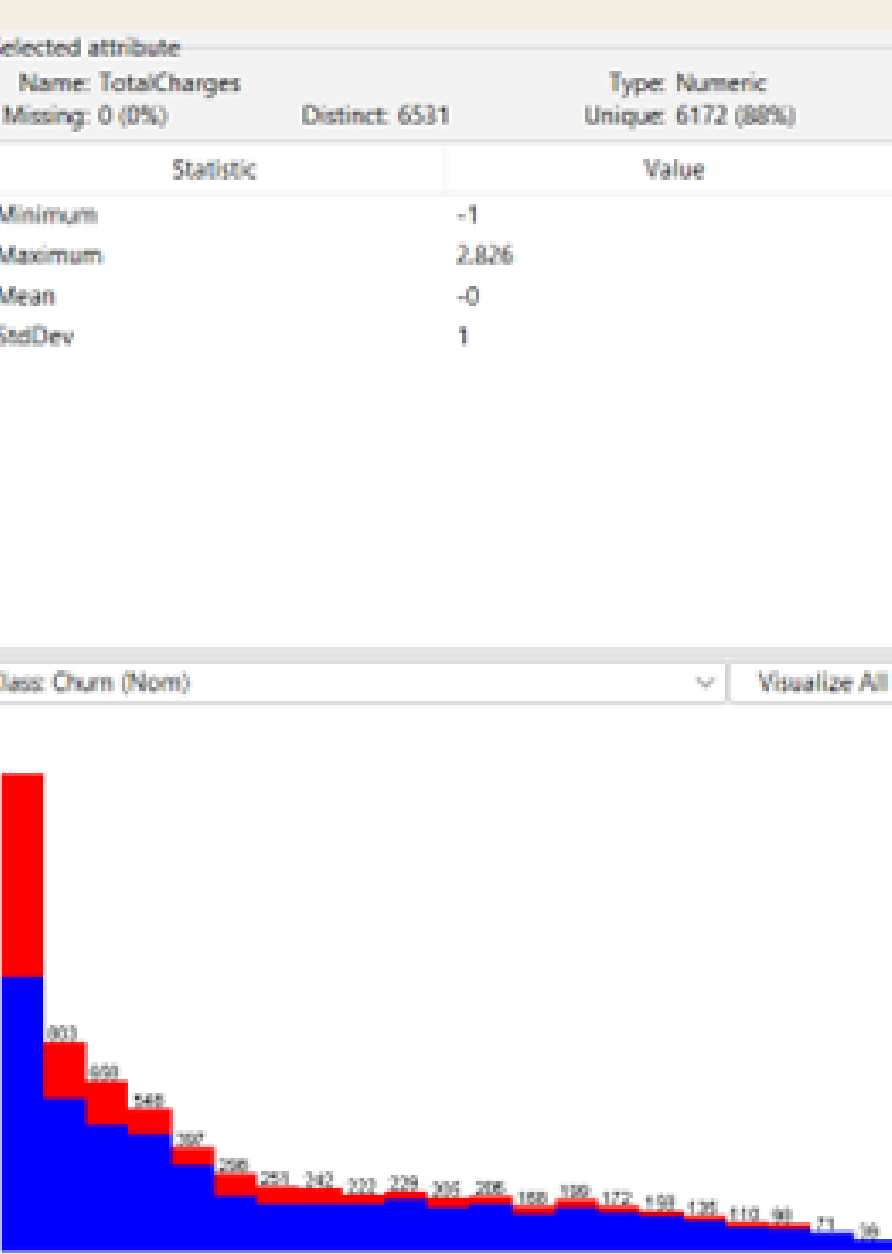
After



Before



After



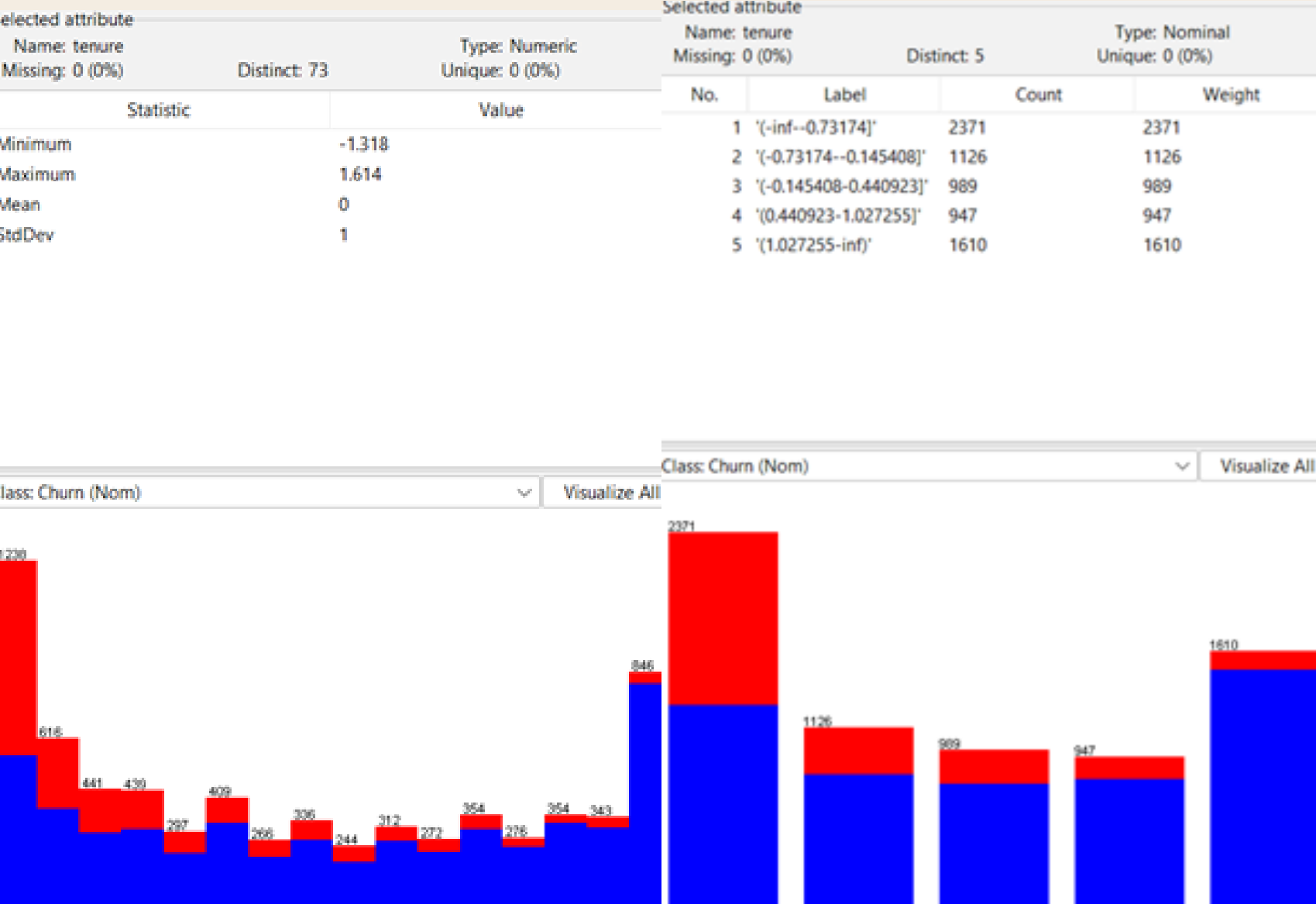
Normalization was applied to the numeric attributes in the dataset because they have different scales and ranges. Z-score normalization was selected and implemented using the Standardize filter in WEKA. This method transforms data to have a mean of zero and a standard deviation of one. Z-score normalization was chosen because it is less sensitive to outliers compared to min-max normalization and is suitable for datasets containing extreme values. The normalization process was applied to the tenure, MonthlyCharges, and TotalCharges attributes only.

DISCRETIZATION

Discretization was applied only to selected numeric attributes:

Before

After



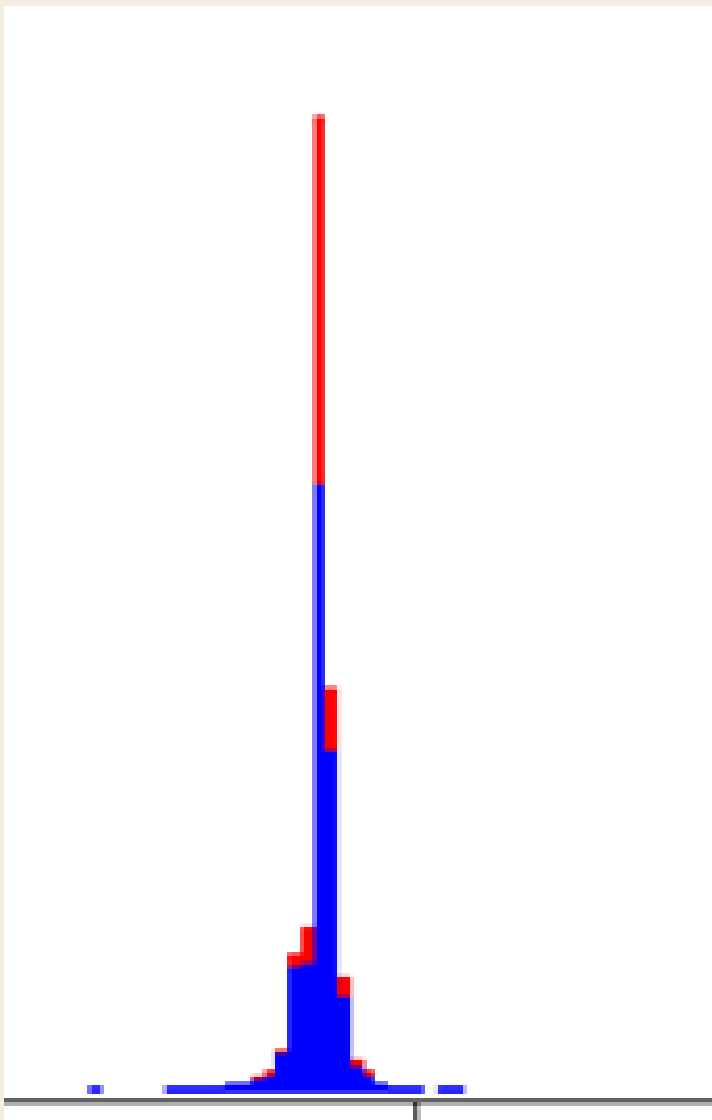
ATTRIBUTE CONSTRUCTION

$$\text{AvgMonthlyCharge} = \text{TotalCharges} / \text{tenure}$$

Weka filter = AddExpression

No.	Name
1	☐ gender
2	☐ SeniorCitizen
3	☐ Partner
4	☐ Dependents
5	☐ tenure
6	☐ PhoneService
7	☐ MultipleLines
8	☐ InternetService
9	☐ OnlineSecurity
10	☐ OnlineBackup
11	☐ DeviceProtection
12	☐ TechSupport
13	☐ StreamingTV
14	☐ StreamingMovies
15	☐ Contract
16	☐ PaperlessBilling
17	☐ PaymentMethod
18	☐ MonthlyCharges
19	☐ TotalCharges
20	☐ Churn

No.	Name
1	☐ gender
2	☐ SeniorCitizen
3	☐ Partner
4	☐ Dependents
5	☐ tenure
6	☐ PhoneService
7	☐ MultipleLines
8	☐ InternetService
9	☐ OnlineSecurity
10	☐ OnlineBackup
11	☐ DeviceProtection
12	☐ TechSupport
13	☐ StreamingTV
14	☐ StreamingMovies
15	☐ Contract
16	☐ PaperlessBilling
17	☐ PaymentMethod
18	☐ MonthlyCharges
19	☐ TotalCharges
20	☐ Churn
21	☒ AvgMonthlyCharge



AvgMonthlyCharge, was constructed by dividing TotalCharges by tenure using the AddExpression filter in WEKA. This attribute represents the average monthly spending of customers and provides better insight into customer billing behaviour. The constructed attribute enhances the dataset by capturing relationships not explicitly represented in the original attributes.

OTHER SPECIFIC PROCESS

Apart from normalization, discretization, and attribute construction, no additional transformation was required. The dataset was already well structured, and all attributes were suitable for classification modelling after the applied preprocessing steps.

B MODEL DEVELOPMENT & EVALUATION

```
Correctly Classified Instances      5482      77.8361 %
Incorrectly Classified Instances    1561      22.1639 %
Kappa statistic                    0.3996
Mean absolute error                0.2756
Root mean squared error            0.4123
Relative absolute error            70.687 %
Root relative squared error        93.3801 %
Total Number of Instances         7043

=== Detailed Accuracy By Class ===

              TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Ar
              0.879   0.501   0.829      0.879   0.854      0.403    0.767    0.861
              0.499   0.121   0.599      0.499   0.544      0.403    0.767    0.511
Weighted Avg.  0.778   0.400   0.768      0.778   0.772      0.403    0.767    0.768

=== Confusion Matrix ===

  a    b  <-- classified as
4549  625 |    a = No
  936  933 |    b = Yes
```

K=10

```
=== Summary ===

Correctly Classified Instances      5488      77.9213 %
Incorrectly Classified Instances    1555      22.0787 %
Kappa statistic                    0.3958
Mean absolute error                0.2772
Root mean squared error            0.4123
Relative absolute error            71.0795 %
Root relative squared error        93.3886 %
Total Number of Instances         7043

=== Detailed Accuracy By Class ===

              TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Are
              0.885   0.514   0.827      0.885   0.855      0.400    0.756    0.847
              0.486   0.115   0.605      0.486   0.539      0.400    0.756    0.514
Weighted Avg.  0.779   0.408   0.768      0.779   0.771      0.400    0.756    0.759

=== Confusion Matrix ===

  a    b  <-- classified as
4580  594 |    a = No
  961  908 |    b = Yes
```

K=20

B MODEL DEVELOPMENT & EVALUATION

```
=== Summary ===

Correctly Classified Instances      1662      78.6559 %
Incorrectly Classified Instances    451      21.3441 %
Kappa statistic                    0.4149
Mean absolute error                 0.2719
Root mean squared error             0.4075
Relative absolute error             70.0284 %
Root relative squared error         93.0077 %
Total Number of Instances          2113

=== Detailed Accuracy By Class ===

              TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Are
              0.883   0.488   0.838     0.883   0.860     0.417   0.766   0.878
              0.512   0.117   0.603     0.512   0.554     0.417   0.766   0.513
Weighted Avg.  0.787   0.392   0.777     0.787   0.781     0.417   0.766   0.784

=== Confusion Matrix ===

  a    b  <-- classified as
1382  184 |    a = No
 267  280 |    b = Yes
```

Psplit(70:30)

```
=== Summary ===

Correctly Classified Instances      1100      78.0696 %
Incorrectly Classified Instances    309      21.9304 %
Kappa statistic                    0.4197
Mean absolute error                 0.2756
Root mean squared error             0.4082
Relative absolute error             70.3933 %
Root relative squared error         91.9681 %
Total Number of Instances          1409

=== Detailed Accuracy By Class ===

              TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Are
              0.874   0.471   0.834     0.874   0.853     0.421   0.763   0.863
              0.529   0.126   0.607     0.529   0.565     0.421   0.763   0.554
Weighted Avg.  0.781   0.378   0.773     0.781   0.776     0.421   0.763   0.780

=== Confusion Matrix ===

  a    b  <-- classified as
 899  130 |    a = No
 179  201 |    b = Yes
```

Psplit(80:20)



REDUCTION

Attribute Evaluator

Choose

CfsSubsetEval -P 1 -E 1

Search Method

Choose

BestFirst -D 1 -N 5

The analysis reduced the dataset from 20 attributes to 5 key attributes: tenure, OnlineSecurity, TechSupport, Contract, and PaymentMethod. These attributes were identified as the most influential factors affecting customer churn. The reduced dataset improves model interpretability and reduces computational complexity.

```
=== Attribute Selection on all input data ===

Search Method:
  Best first.
  Start set: no attributes
  Search direction: forward
  Stale search after 5 node expansions
  Total number of subsets evaluated: 138
  Merit of best subset found:      0.141

Attribute Subset Evaluator (supervised, Class (nominal): 20 Churn):
  CFS Subset Evaluator
  Including locally predictive attributes

Selected attributes: 5,9,12,15,17 : 5
      tenure
      OnlineSecurity
      TechSupport
      Contract
      PaymentMethod
```

No.		Name
1	☒	tenure
2	☐	OnlineSecurity
3	☐	TechSupport
4	☐	Contract
5	☐	PaymentMethod
6	☐	AvgMonthlyCharge
7	☐	Churn

REDO

```
=== Summary ===

Correctly Classified Instances      5607      79.611 %
Incorrectly Classified Instances    1436      20.389 %
Kappa statistic                    0.4261
Mean absolute error                 0.2885
Root mean squared error             0.3826
Relative absolute error             73.9773 %
Root relative squared error         86.6423 %
Total Number of Instances          7043

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC A
                0.911    0.523    0.828     0.911    0.868     0.436    0.803    0.909
                0.477    0.089    0.660     0.477    0.554     0.436    0.803    0.569
Weighted Avg.   0.796    0.408    0.784     0.796    0.785     0.436    0.803    0.819

=== Confusion Matrix ===

      a    b  <-- classified as
4716  458 |    a = No
  978  891 |    b = Yes
```

K=10

```
=== Summary ===

Correctly Classified Instances      5609      79.6394 %
Incorrectly Classified Instances    1434      20.3606 %
Kappa statistic                    0.4277
Mean absolute error                 0.2881
Root mean squared error             0.3818
Relative absolute error             73.8872 %
Root relative squared error         86.4707 %
Total Number of Instances          7043

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC A
                0.911    0.521    0.829     0.911    0.868     0.437    0.801    0.907
                0.479    0.089    0.660     0.479    0.555     0.437    0.801    0.564
Weighted Avg.   0.796    0.406    0.784     0.796    0.785     0.437    0.801    0.816

=== Confusion Matrix ===

      a    b  <-- classified as
4713  461 |    a = No
  973  896 |    b = Yes
```

K=20

```
=== Summary ===

Correctly Classified Instances      1677           79.3658 %
Incorrectly Classified Instances    436           20.6342 %
Kappa statistic                    0.4149
Mean absolute error                0.2844
Root mean squared error            0.3838
Relative absolute error            73.2618 %
Root relative squared error        87.6114 %
Total Number of Instances          2113

=== Detailed Accuracy By Class ===

               TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area
Weighted Avg.   0.904   0.523   0.832     0.904   0.867     0.422   0.803   0.907
               0.477   0.096   0.635     0.477   0.545     0.422   0.803   0.550
Weighted Avg.   0.794   0.412   0.781     0.794   0.783     0.422   0.803   0.815

=== Confusion Matrix ===

   a    b  <-- classified as
1416  150 |    a = No
 286   261 |    b = Yes
```

Psplit 70

```
=== Summary ===

Correctly Classified Instances      1125           79.8439 %
Incorrectly Classified Instances    284           20.1561 %
Kappa statistic                    0.443
Mean absolute error                0.292
Root mean squared error            0.3848
Relative absolute error            74.5897 %
Root relative squared error        86.6882 %
Total Number of Instances          1409

=== Detailed Accuracy By Class ===

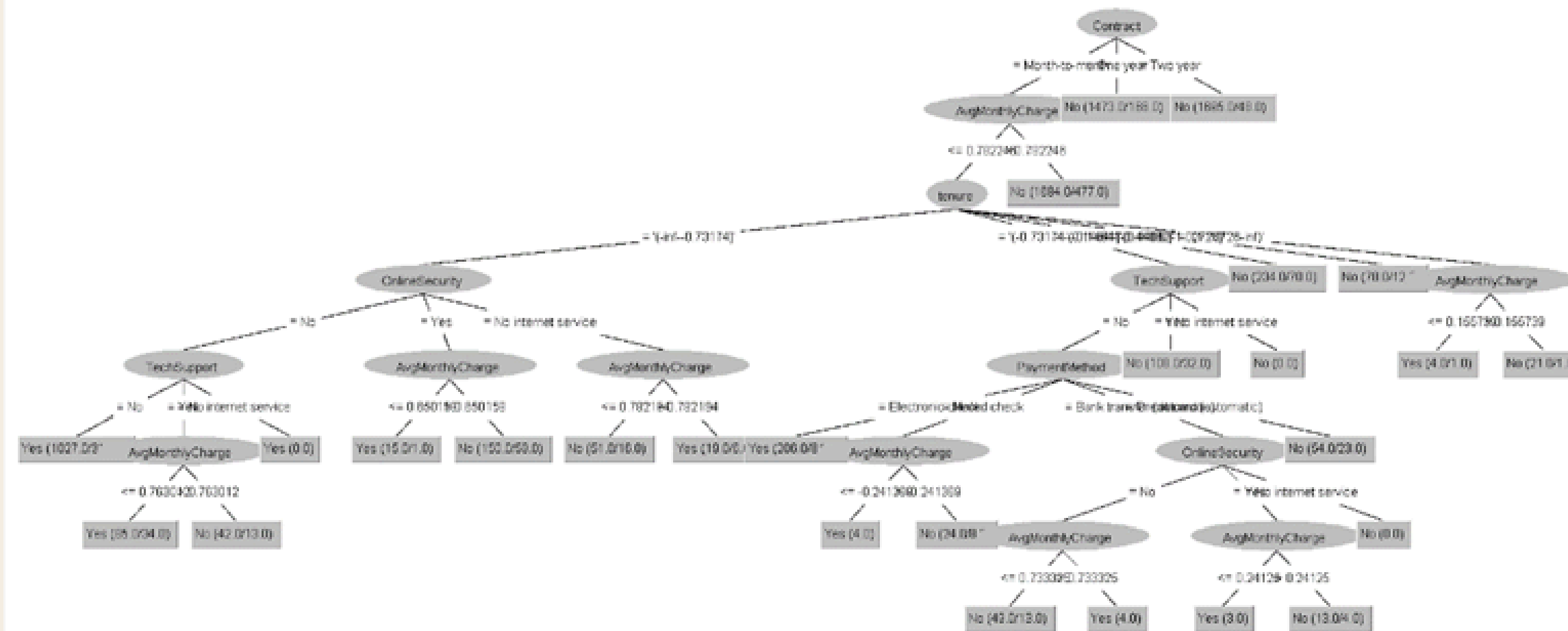
               TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area
Weighted Avg.   0.910   0.503   0.831     0.910   0.868     0.451   0.807   0.901
               0.497   0.090   0.670     0.497   0.571     0.451   0.807   0.576
Weighted Avg.   0.798   0.391   0.787     0.798   0.788     0.451   0.807   0.813

=== Confusion Matrix ===

   a    b  <-- classified as
 936   93 |    a = No
 191  189 |    b = Yes
```

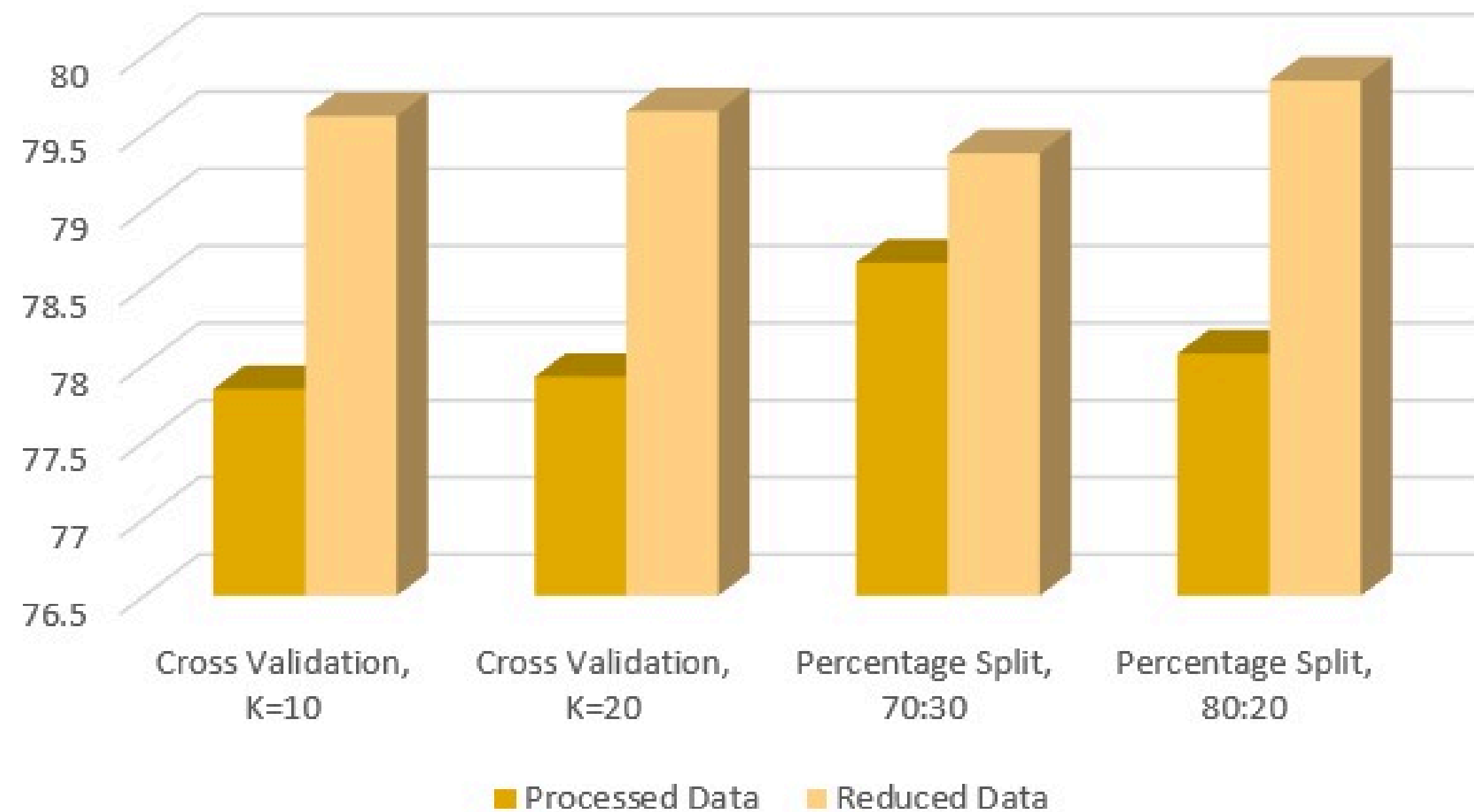
Psplit 80

RESULT



GRAPH

Comparison of Accuracy between Processed Data and Reduced Data



CONCLUSION

The reduced dataset achieved higher accuracy across all tests, indicating that attribute reduction improved model performance. This improvement occurred because irrelevant and redundant attributes were removed, allowing the decision tree to focus on the most significant features influencing customer churn, such as tenure, contract type, online security, technical support, and payment method. As a result, noise and overfitting were reduced, leading to better generalization. Among all evaluation methods, the 80:20 percentage split produced the highest accuracy due to the larger training dataset, which enabled the model to learn more representative customer behaviour patterns while maintaining sufficient data for reliable testing.

GROUP 1

THANK YOU

ISP565: Data Mining

Presented by Group 1